

Stacking-Driven Explainable Machine Learning Method: Uncovering Anxiety Risk in Elderly Women under Home-based Care from CLHLS

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Authors' contributions: Kai Wang conducted model building, experiment and result analysis, and paper writing.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics approval and consent to participate

The studies involving humans were approved by The Biomedical Ethics Review Committee of Peking University (Beijing, China). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

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Author contributions

Kai Wang: Formal analysis, Supervision, Writing – review & editing. Yu Du: Writing – original draft, Writing – review & editing.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Supplementary material

The Supplementary material for this article can be found online.

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Abstract

Background

Amidst accelerating global population aging, elderly women receiving home-based care face elevated risks of anxiety, necessitating precise predictive tools to improve mental health outcomes. This study aims to identify and develop an interpretable prediction model for quantifying the likelihood of anxious symptoms in this vulnerable population in China.

Methods

Model development and internal validation utilized data from the fifth wave of the China Longitudinal Health Longevity Survey (CLHLS, 2008), involving 2,313 participants aged ≥ 65 years under home-based care. 40 variables were defined based on the biopsychosocial model framework. Key predictors were selected via LASSO regression, followed by the integration of three data-balancing techniques. A Stacking ensemble model incorporating six machine learning algorithms was constructed to predict anxiety disorder. Shapley Additive exPlanations (SHAP) were employed to elucidate feature importance and model interpretability.

Results

Anxiety disorder prevalence was notable among home-based elderly women. After data balancing, the stacking-driven interpretable ensemble model demonstrated superior performance, achieving high area under the receiver operating characteristic curve (AUROC: 0.833) and area under the precision-recall curve (AUPRC: 0.818), significantly outperforming single-algorithm benchmarks. SHAP analysis identified critical risk factors spanning health status (e.g., self-rated health, chronic pain), lifestyle behaviors (e.g., sleep duration, physical activity), and social determinants (e.g., social support, financial stability).

Conclusion

The Stacking-driven interpretable ensemble model demonstrated superior predictive performance (AUC=0.818) compared to single-algorithm approaches, effectively integrating multidimensional risk factors from biological, psychological, and social domains. SHAP analysis identified body pain severity, sleep duration, self-rated health, and environmental factors (air quality satisfaction, hearing impairment) as key predictors, providing actionable insights for targeted interventions. This model enables community healthcare providers to stratify elderly women under home-based care into risk tiers, facilitating early psychological assessment and personalized support strategies. The integration of machine learning with biopsychosocial theory offers a paradigm shift in geriatric mental health management, emphasizing the necessity of multidimensional intervention frameworks.

Limitation

This study has three primary limitations: (1) Reliance on cross-sectional CLHLS 2008 data restricts longitudinal validation of risk factor dynamics; (2) Self-reported GAD-7 measurements may introduce recall bias and underdiagnoses of subclinical cases; (3) LASSO-based feature selection potentially excludes clinically relevant variables with weak statistical associations, such as financial strain and caregiving intensity. Future iterations should incorporate longitudinal biomarkers and socioeconomic stressors to enhance clinical translatability.

Keywords: anxiety risk; China; women under home-based care; machine learning; prediction model

1. Background

Late-life anxiety disorder represents a critical public health challenge, particularly among elderly women receiving home-based care, who face 1.5-2 times higher anxiety risks post-menopause due to estrogen decline and social isolation linked to spousal loss or rural youth migration (Wei et al., 2019). With 22% of the global population projected to be over 60 by 2050 (Wang et al., 2022), the expanding home-care demographic underscores an urgent need for precise risk prediction tools. Traditional models inadequately address multifactorial interactions between physiological vulnerabilities (e.g., neurotransmitter disruptions from hormonal changes), psychosocial stressors (30% prevalence of chronic loneliness) (Cui et al., 2024), and caregiving burdens (40% anxiety rates in long-term family caregivers) (Diaz-Gallo et al., 2021). These limitations hinder early intervention strategies to mitigate anxious cascading impacts on functional decline, familial strain, and healthcare costs. Thus, developing an interpretable prediction model that can precisely estimate the likelihood of anxious symptoms in elderly women under home-based care is an urgent need in the current field of geriatric mental health research.

The realm of predicting anxiety in elderly women, risk prediction models has been developed in previous studies (Tan et al., 2023; Liet al., 2025). However, these existing models have several shortcomings. Firstly, many of them utilize a limited number of risk factors. For example, some studies only consider one or two aspects such as physical health or social support, neglecting other important factors like psychological resilience and lifestyle habits (Kim et al., 2021). This narrow focus can lead to inaccurate predictions, as anxiety is a complex disorder influenced by multiple elements. Secondly, most previous models do not fully account for the complex interactions among different factors. In reality, physiological, psychological, and social factors interact intricately, and ignoring these interactions can lead to an incomplete understanding of the anxiety risk mechanism (Guo et al., 2021). Additionally, few studies have specifically focused on the unique characteristics of elderly women under home-based care in China, such as the influence of traditional cultural values on their mental health (Wang et al., 2023). Given these limitations, there is a clear gap in the existing research, and it is necessary to conduct a more in - depth study to fill this void.

Machine learning has emerged as a transformative force in the realm of disease risk prediction, offering unparalleled potential to revolutionize healthcare decision-making. Its ability to analyze complex patterns within large datasets has proven to be a game - changer in this field (Ehtemam et al., 2024). One of the most significant contributions of machine learning is its capacity to capture non - linear relationships among variables (Vu et al., 2025). While existing methodologies employ machine learning for anxiety risk prediction, two critical methodological limitations persist. First, current predictive approaches predominantly rely on single-algorithm frameworks (e.g., logistic regression or random forest alone), constraining computational precision through inadequate integration of complementary learners' strengths (Qiu et al., 2024). Second, insufficient attention has been given to the inherent limitations of cross-sectional survey data, particularly regarding missing key variables and data imbalance - a common challenge in epidemiological studies like China Health and Retirement Longitudinal Study (CLHLS) (Wang et al., 2024). Existing studies lack comprehensive mechanistic analyses of how different balancing techniques (e.g., SMOTE vs. ADASYN) interact with feature selection processes to influence model performance, resulting in prediction outcomes with compromised interpretability (Zheng et al., 2024). This dual deficiency in algorithmic diversity and data preprocessing transparency substantially hinders clinical applicability, particularly when handling multidimensional geriatric datasets encompassing biological, psychological, and social determinants.

In summary, this study focuses on elderly women receiving home-based care in China. By using data from the 2008 CLHLS cohort, we attempted to define variables conducted by LASSO

regression for feature selection based on the Bio-Psycho-Social model, and apply three data -balancing techniques to construct a stacking model with five machine-learning algorithms. Finally, SHAP analysis was utilized to identify feature importance, striving to develop a more accurate and interpretable anxiety -risk prediction model for early intervention and mental health improvement.

2. Method

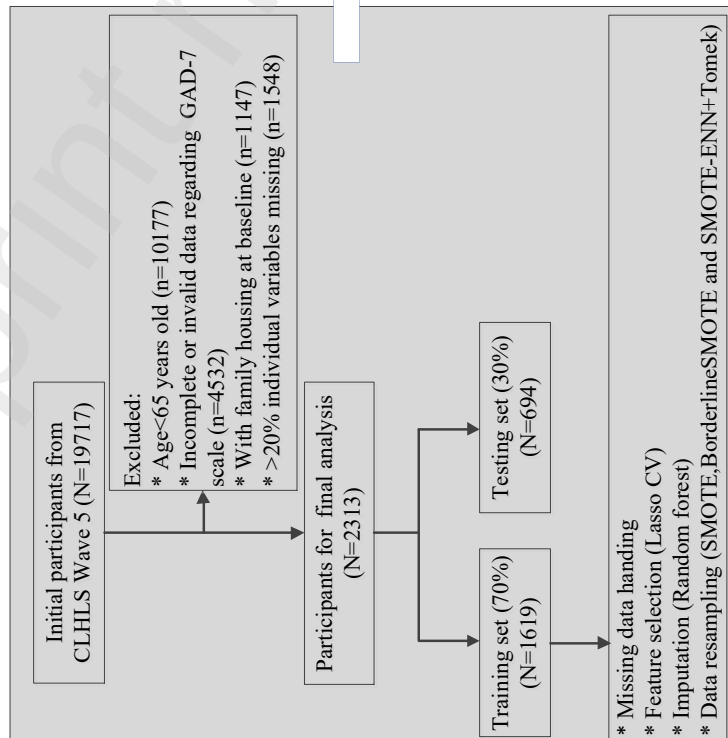
2.1 Data sources

The China Longitudinal Health Longevity Survey (CLHLS) aims to collect a set of high-quality microdata representing households and individuals of Chinese people, which regularly conducts follow-up surveys on the elderly aged 65 and above in 23 provinces and cities across the country, covering health status, economic background, lifestyle and other dimensions. As one of the largest follow-up surveys of the elderly in the world, its data provide important support for the study of healthy aging, longevity mechanisms and related policy formulation. The baseline data and follow-up data of CLHLS were obtained on the basis of approval from the Ethics Review Committee of Peking University (IRB00001052-13074), and all respondents signed the informed consent form. The data used in this study comes from wave 5 of CLHLS in 2008 (<http://CLHLS.pku.edu.cn/en>). A total of 19,717 participants were enrolled and were screened to exclude subjects who (1) were younger than 65 years of age, (2) were male or female without receiving home-based care, (3) had missing information on the Generalized Anxiety Disorder 7-item Scale (GAD-7), and (4) had invalid or missing individual variables with more than 20%. After screening, a total of 2,313 eligible subjects were enrolled in the baseline study as shown in Fig.1.

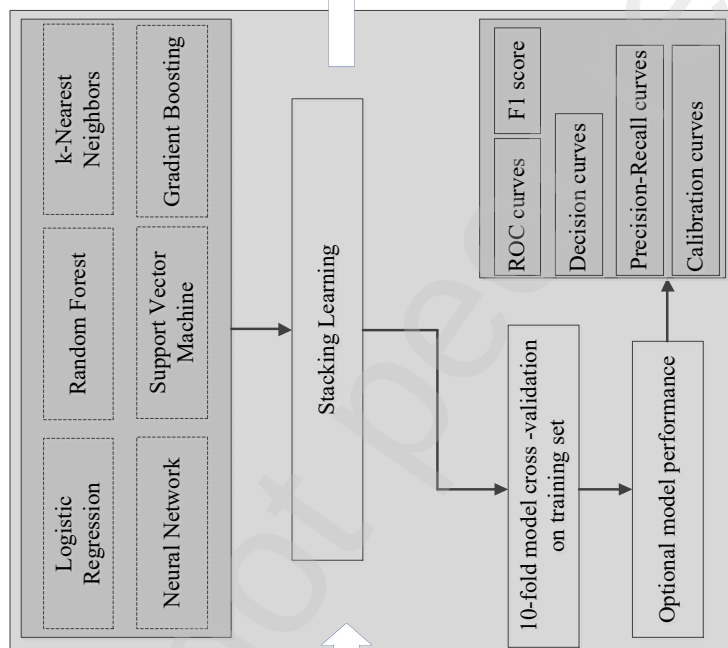
2.2 Predictor variables

Based on three aspects of the Bio-Psycho-Social model (Delshad et al., 2022), combined with the clinical significance and scientific knowledge of anxiety (Tan et al., 2023; An et al., 2024), and predictors identified in previous publications (Han et al., 2023; Ai et al., 2024), we considered a range of key candidate variables associated with psychological and social factors. These variables are mainly classified into three main categories: biological function, psychological condition, and social function. (1) The biological function data included self -rated health status, chronic disease total, body pain level, received inpatient care, eyesight problem, hearing problem, blood pressure or sugar problem, lost teeth, sleep duration, physical activities, falling down and fractured hip, BADL disability (Basic activities of daily living), IADL disability (Instrumental activities of daily living). Specifically, BADL disability is mainly assessed for the elderly's most basic self-care activities, including eating, dressing, bathing, using the toilet, and moving around indoors. IADL disability focuses on measuring the elderly's ability to participate in more complex daily activities, including using the phone, shopping, cooking, and managing finances. This paper uses the Katz weighted index to determine the difficulty level of the elderly in performing the above activities, that is, the presence of any behavioral disorder is judged as BADL or IADL disability. (2) The psychological condition data included MMSE status (Mini mental state exam, Assess overall cognitive function), TICS status (Interview for cognitive status), WR status (Word recall, Assess memory), RF status (Retrieval fluency, Assess language and cognitive fluency), and CSI-D status (Community screening instrument for dementia). (3) The social function data included gender, age, nation, religious beliefs, address type, marital status, educational level, residential district, building type, steps to household's flat main entrance, government pension, and amount of alcohol intake, cigarettes consumption, social activities, Internet entertainment, agricultural work, fringe benefits and pension type. Detailed information regarding predictor variables is provided in supplementary Table S1.

Data Screen



Stacking-Driven Ensemble Learning



Visualization and Prediction

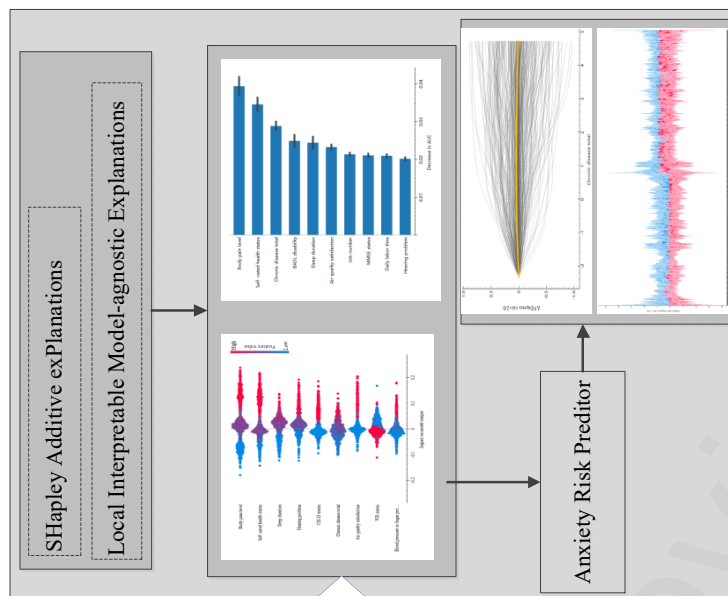


Fig.1 Stacking-Driven Machine learning flow for this study

2.3 Outcome definition

The outcome variable of this study is anxious symptoms. Anxious symptoms were measured using the GAD-7 scale (Generalized Anxiety Disorder 7-item Scale), which is a concise and effective psychological measurement tool specifically developed for assessing anxious symptoms in Chinese elderly people (Wang et al., 2022; Wang et al., 2023). GAD-7 utilizes a seven-level scoring system from 0 to 3 points. A score of “0” indicates that the symptom has never occurred in the past week; “3” indicates that the symptom appears almost every day, severely affecting the elderly's daily life and mental health. The total score is calculated by summing the 7 item scores, with higher scores indicating greater severity of anxious symptoms in elderly individuals. Generally speaking, if the total score exceeds 5 points, it is defined as clinically significant anxious symptoms, which require further assessment and intervention. The Cronbach coefficient of this scale is 0.919.

2.4 Data preprocessing and feature engineering

In this study, we stratified the data according to the results of anxious symptoms and randomly divided the dataset into a training set and a test set at a ratio of 7:3 to avoid overfitting. To handle missing values, we employed the random forest imputation method. This method constructs multiple decision trees using the complete feature variables in the data and predicts the missing values based on other non - missing features. Compared with traditional imputation methods (such as mode imputation or mean imputation), random forest imputation can better capture the internal patterns of the data through the comprehensive prediction of multiple decision trees, performs better when dealing with complex datasets, and effectively reduces the bias introduced by imputation (Ouet et al., 2024).

In addition, feature selection is a crucial step in determining the effectiveness of model training. It can not only reduce the data dimension, decrease the computational load, and improve the speed of model training and prediction but also remove redundant and irrelevant features, enhancing the generalization ability of the model (Chen et al., 2024). To effectively reduce the risk of data leakage, we strictly confined feature selection to the training set. Specifically, we used the Least Absolute Shrinkage and Selection Operator (Lasso CV) and selected key variables with non - zero coefficients through 10 - fold cross - validation (Tang et al., 2021).

To effectively address class imbalance in the dataset, the Synthetic Minority Over - sampling Technique (SMOTE) has been widely used to handle highly skewed datasets, thereby improving the model's classification performance on minority class samples (Nguyen et al., 2023). In addition, methods such as BorderlineSMOTE (AlMahadin et al., 2023) and Synthetic minority over - sampling technique, edited nearest neighbors and tomes Links (SMOTE - ENN+Tomek) (Alsinglawi et al., 2022) have also been proposed successively. These methods enhance the quality of oversampling from different perspectives by identifying boundary samples, removing noise and redundant samples, etc. Among them, BorderlineSMOTE can focus on the boundary samples of the minority class, effectively avoiding the generation of too many unnecessary samples in areas far from the classification boundary and significantly improving the model's classification performance in the boundary area. SMOTE - ENN+Tomek is a method that combines the SMOTE algorithm with ENN (Edited Nearest Neighbors) and Tomek Links. This method increases the number of minority class samples through SMOTE oversampling, removes noise through ENN, and eliminates redundant samples through Tomek Links to optimize the sample distribution and improve the model's classification effect when dealing with imbalanced datasets.

2.5 Model construction for risk prediction

To obtain the optimal prediction model, we first employed five machine learning methods and an ensemble learning algorithm consisting of five base models to train the meta - model. We

also attempted to combine the above - mentioned different machine learning methods with three resampling techniques to test and analyze the model's performance. Specifically, six machine learning and ensemble learning algorithms were used to develop the prediction models, namely Logistic Regression (LR), k - Nearest Neighbors (kNN), Random Forest (RF), Support Vector Machine (SVM), Gradient Boosting (GB), and Neural Network (NN). Additionally, Stacking uses the prediction results of the above five algorithms as new inputs and enhances the overall performance through a hierarchical learning mechanism. This process is mainly divided into two stages. Stage 1 (Base model training): The training set is divided into five subsets using 10 - fold cross - validation. Each base model (such as SVM, neural network, etc.) is trained on nine folds of the data and generates prediction results (referred to as Out - of - Fold Predictions) on the remaining fold to avoid data leakage. Stage 2 (Meta - model training): The prediction results are used as new features and input into the meta - model for training. The meta - model obtains the final prediction set based on the prediction results of the base models.

LR is a supervised algorithm based on linear regression that predicts classification probabilities through a logistic function. kNN is an instance - based non - parametric classification algorithm that predicts new samples based on the categories of the nearest k neighbors. RF is an ensemble algorithm based on decision trees, which randomly constructs multiple trees and votes/averages the results. SVM is an algorithm based on statistical learning theory that searches for the optimal hyperplane for classification and regression. NN is a computational model that mimics biological neural networks and can handle complex non - linear relationships. Table S2 lists the optimal hyperparameter settings for each model.

2.6 Model evaluation and interpretation

To evaluate the model performance, we calculated classification metrics on the test set, including the area under the receiver operating characteristic curve (AUC), accuracy, recall, and F1 - score. Additionally, although the Stacking - Driven Learning method can effectively integrate the advantages of different models and leverage their diversity and complementarity to improve the overall classification performance, due to the fact that the input of the meta - model relies on a hierarchical learning mechanism, the interpretability of this model is poor. The black - box nature of the meta - model makes it difficult to understand the final prediction process. Therefore, we used the SHapley Additive exPlanations (SHAP) method for model interpretability analysis. This method reveals the influence degree of different features on the prediction results by calculating the average individual contribution value of each feature and plotting the feature contribution bar chart and summary chart (Wang et al., 2021). Moreover, to explain in detail the prediction mechanism of anxious symptoms in elderly women receiving home - based care, we further developed a prediction model based on the optimal machine learning method. By changing the values of the target feature within a certain range while keeping other features fixed, we obtained the prediction results under different values of the target feature and then plotted the Individual Conditional Expectation Curve (ICEC).

2.7 Statistical analysis

All variables in this study were categorized and presented as frequencies and percentages. Comparisons of count data between groups were conducted using the chi - square test, while comparisons of measurement data between groups were performed using the t - test to evaluate the relationship between predictor variables and anxious symptoms, with the specific method depending on the data distribution. All analyses were carried out using R version 4.3.0, and a P - value less than 0.05 was considered statistically significant. Selected characteristics of CLHLS participants are shown in Table S2 in Appendix 1.

3. Result

3.1 Geographical distribution of anxiety risk

Figure 2 presents the prevalence of anxiety risk in China, based on data from the wave 5 of CLHLS conducted in 2008. Hainan, Shanxi, Chongqing, Heilongjiang Jiangxi and Anhui provinces reported a high prevalence of anxiety risk, with rates exceeding 20% (16/57, 28.1%; 6/22, 27.3%; 19/84, 22.6%, 7/34, 20.6%, 10/49, 20.4%; 5/25, 20.0%, respectively). In contrast, Shandong, Shanghai, Jilin and Liaoning had a low prevalence of anxiety risk, with rates below 10% (31/359, 8.6%; 2/24, 8.3%; 1/15, 6.7%; 3/54, 5.6%;). The remaining provinces exhibited a moderate prevalence of anxiety risk, ranging from 10% to 20%. Detailed information regarding the prevalence of anxiety risk in each province is presented in Table S3 in Appendix 1.

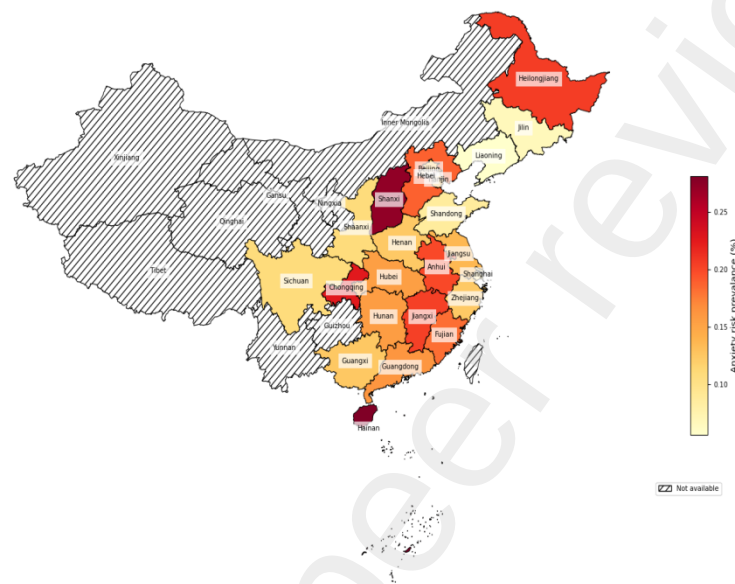


Figure 2. The prevalence of anxiety risk in China from CLHLS of wave 5.

3.2 Baseline characteristics of anxiety risk in elderly women receiving home-based care

This study included a total of 2,313 participants, consisting of 962 elderly women living in urban areas and 1,351 living in rural areas or urban - rural fringe areas. Among the participants, 382 had anxious symptoms with a GAD-7 score exceeding 5, and the prevalence rate was 16.5%. Among the 40 variables included in the study, the following variables were significantly associated with anxious symptoms: Age level, Self - rated health status, Chronic disease total, Body pain level, Eyesight problem, Hearing problem, Blood pressure or Sugar problem, BADL disability, IADL disability, Sleep duration, Falling down, Fractured hip, Amount of alcohol intake, Cigarettes consumption, MMSE status, TICS status, WR status, RF status, Children satisfaction, Marriage satisfaction, Religious beliefs, Address type, Government Pension, Marital status, Educational Level, Residential district, Building type, Steps to household's flat main entrance, Agricultural work, Fringe benefits, Pension type, Household debts, Household type, and House elevator.

3.3 Identification of predictive factors

In the training dataset, the optimal parameters of the Lasso CV method were selected through ten - fold cross - validation (with a lambda value of 0.0134). We finally identified 20 key variables with non - zero coefficients: Age level, Self - rated health status, Chronic disease total, Body pain level, Eyesight problem, Hearing problem, Blood pressure or Sugar problem, BADL disability, IADL disability, Sleep duration, MMSE status, TICS status, WR status, RF status, Children satisfaction, Marriage satisfaction, Religious beliefs, Government Pension, Marital status, and

3.4 Construction and evaluation of the ensemble learning model

A Stacking ensemble model was constructed using six machine learning methods, and it was combined with three different data balancing methods to build a risk prediction model for anxious symptoms in elderly women receiving home-based care. Best parameters for predicting anxious disorder on 10-fold grid search are exhibited in Table S4 in Appendix 1. The ROC curve and Calibration Curve on the test set are shown in Figure 3 and Figure 4 respectively. The results indicate that the three different data balancing methods have significant differential impacts on the prediction performance of the classifiers. Additionally, Stacking-SET and GB-SET are closer to the diagonal in the high prediction probability interval (such as 0.6–1.0), showing relatively good calibration, while some models (such as SVM-SET) deviate more obviously as a whole, especially in the low prediction probability interval (0–0.4), with a large deviation between the predicted probability and the actual probability.

The Stacking-SET method achieved the best accuracy in predicting positive samples. Moreover, among the single models, the kNN-SET method demonstrated the best prediction and classification performance (AUC = 0.818) among all those using the SMOTE-ENN+Tomek data balancing method. It is not only higher than the best classifier GB-SMOTE (AUC = 0.762) using the SMOTE balancing method but also higher than the best classifier GB-BSM (AUC = 0.757) using the BorderlineSMOTE balancing method. However, GB has a more stable comprehensive performance among different data balancing methods, with an average AUC of 0.776 under the three methods. Table S3 shows the complete performance indicators, including AUC, accuracy, recall, F1-score, and MCC. The Precision-Recall Curve and Lift Curve on the test set are shown in Figure 5. Overall, the indicators of the six base classifiers have all improved when combined with the SMOTE-ENN+Tomek data balancing method. SVM has shown the greatest improvement in five indicators: AUC (SVM-BSM: 0.555, SVM-SET: 0.777), CA (SVM-SMOTE: 0.537, SVM-SET: 0.743), F1 (SVM-SMOTE: 0.512, SVM-SET: 0.663), and recall (SVM-SMOTE: 0.483, SVM-SET: 0.760). The above results indicate that the SVM method is highly affected by different types of data balancing techniques and has great volatility. In addition, kNN has the greatest improvement in accuracy (kNN-BSM: 0.609, kNN-SET: 0.701). Additional information regarding the results for balancing methods on six machine leanings exhibited in Table S5 in Appendix 1.

The associations between different features and anxiety may vary due to differences in the elderly care regions. Considering the characteristics of urban and rural development in China, we combined the urban-rural fringe areas and rural areas and collectively referred to them as the large rural areas. Therefore, we conducted grouped predictions for participants living in different regions separately. In urban areas and large rural areas, the Stacking-SET method still performed excellently (with AUC values of 0.793 and 0.786 respectively).

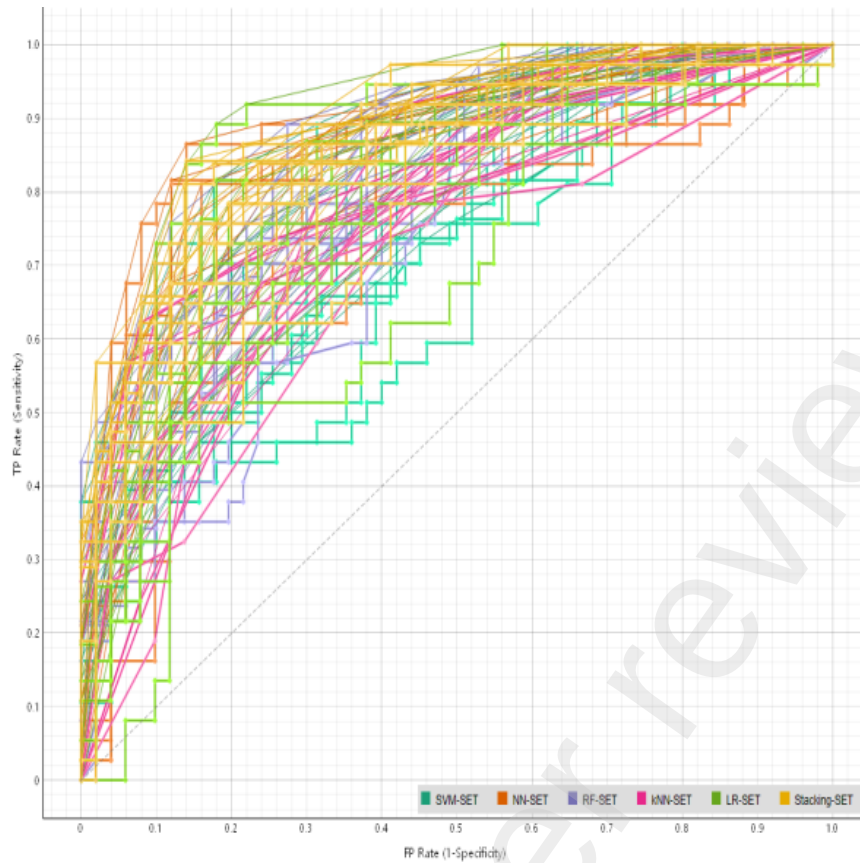


Figure 3. The ROC curve generated on the test dataset.

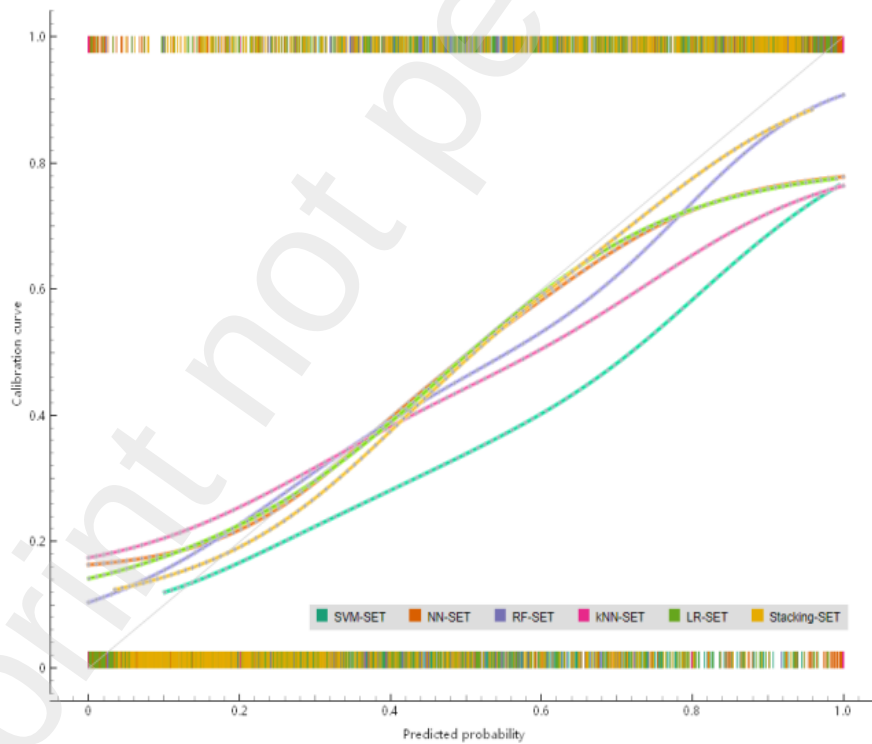


Figure 4. Calibration curve generated on the test dataset.

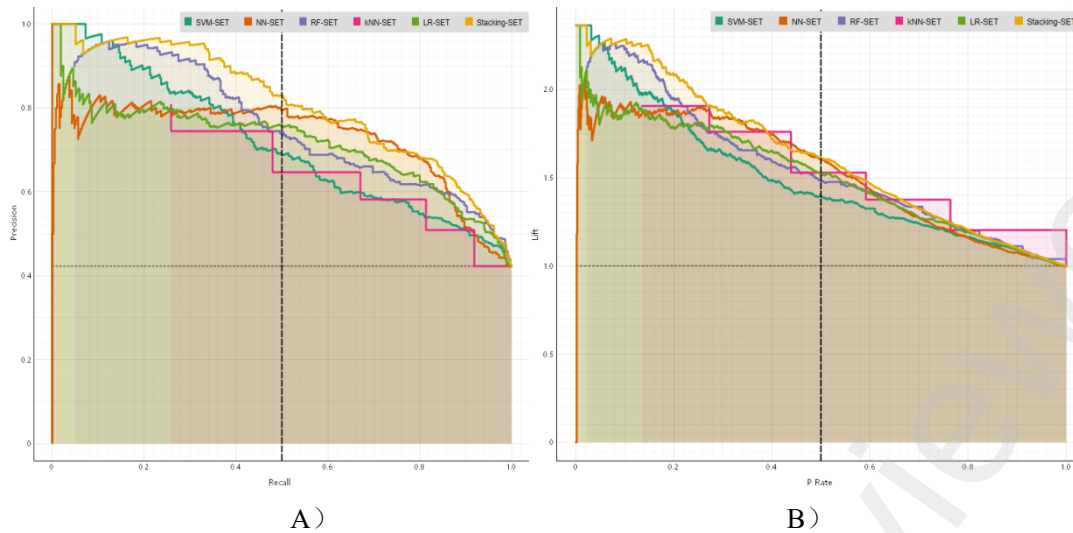


Figure 5. Evaluation index curve on test set. Note: A) Accuracy and recall curve on the test dataset; B) Lift curve generated on the test dataset

3.5 Model interpretation

We calculated and ranked the average SHAP values of the Stacking-SET model to identify important predictive variables and explain their impact on the risk of anxious symptoms in different groups. Figure 6 shows the top 10 features that are most important for the prediction results in the entire target population. Body pain level ranked first, followed by Air quality satisfaction, sleep duration, self-rated health status, and hearing problem. The top features in the SHAP summary plot indicate a more significant impact on the prediction. Body pain level still ranked first. From the color distribution, the red area (representing "high feature values") is concentrated in the positive impact area, indicating that when an individual has a high level of body pain, the positive impact on the model output is more significant. That is, the more severe the pain, the more likely the model is to predict a anxious risk. The last-ranked feature is Chronic disease total. When it is blue (low feature value), it mostly concentrates in the area close to 0 or negative values on the abscissa ("impact on the model output"). This suggests that when an individual has a small total number of chronic diseases, the positive driving effect on the model's prediction of anxious symptoms is weak, and it may even slightly reduce the probability that the model predicts as the target category. It should be noted that features with lower rankings contribute less to anxious symptoms but can still explain the anxious risk of some individuals.

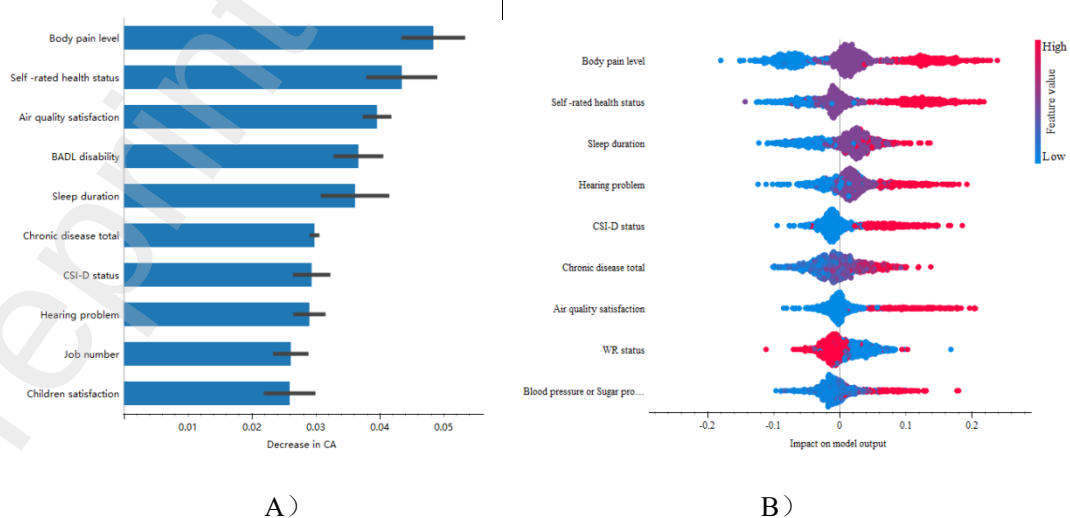


Figure 6. Shapley Additive explanation (SHAP) values and Bar chat of variable contributions on absolute values of SHAP.

To deeply validate the influence mechanism of core features on the target category, Figure 7

presents the Individual Conditional Expectation (ICE) of the variable "Body pain level", which has the strongest impact on anxious symptoms, to analyze the degree of influence of features on the model's prediction results. Specifically, by demonstrating the influence of "Body pain level" and "Self - rated health status" on the prediction probability of the target category, we verified the conclusion that body pain level and self-rated health status are the core driving features for predicting anxious risk. In the figure, the red curve (the target category of 2.0 indicates an anxious risk) shows that the higher the degree of body pain and the lower the self - rated health assessment, the higher the probability that the model predicts a anxious risk. Compared with the non - anxious group (category 1.0), the group at risk of anxiety is more sensitive to the degree of pain and the level of self - rated health assessment. On the other hand, the ICE lines of "Daily labor time" and "Chronic disease total" are more divergent, reflecting significant heterogeneity in individual responses and the complexity of the feature action mechanism. Taking daily labor time as an example, its impact on anxious risk varies highly. For some samples, as the daily labor time increases, the probability of anxious risk rises significantly (the red line goes up), possibly because excessive labor leads to fatigue and stress accumulation, increasing the risk of anxiety. However, some samples show the opposite trend (the blue line goes down), suggesting that some individuals may obtain a sense of achievement or social connection through labor, thereby reducing the risk of anxiety. This divergence reflects that the relationship between "labor time" and anxious risk is not a simple causal one but is regulated by multiple factors such as the nature of labor (e.g., voluntary vs. forced labor), individual psychological adjustment ability, and the labor environment.

In addition, in the classification prediction of high-risk populations, the conditional expectations of the 10 important features are all higher than those of non - important features, indicating that the prediction of anxious risk is the result of the interaction of multi - dimensional features. The above results further validate the correctness of the SHAP analysis. Moreover, the non - zero conditional expectations of multiple feature variables (indicating a certain contribution) can explain the risk level of certain individuals.

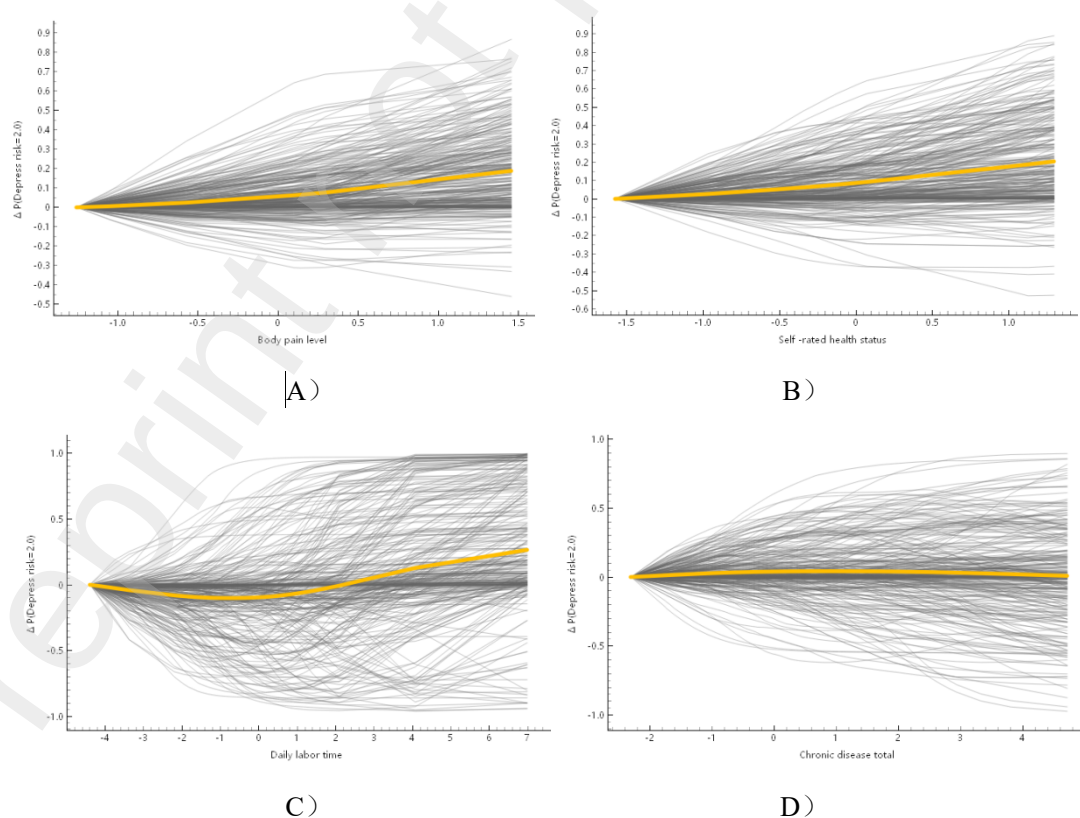


Figure 7. ICE values for four variable characteristics on anxiety risk. Note A) ICE curve for body pain level; B) ICE curve for self-rated health status; C) ICE curve for daily labor time; D) ICE

curve for chronic disease total.

To further illustrate the explanatory role of characteristic variables of different importance in developing anxious risk in individuals, we calculate LIME values (Local Interpretable Model-agnostic Explanations) for 10 important features in each sample and analyze the degree of influence of the model on the output. Figure 8 is the calculated result of the Body pain level. It can be seen that the blue area (which may represent a positive impact) and the red area (which may represent a negative impact) are relatively concentrated and regular. The blue area is relatively stable at the higher position of the longitudinal axis, and the red area is also relatively stable at the lower position of the longitudinal axis, indicating that after being ranked by output, the influence of the degree of physical pain on the prediction of depression risk shows a certain commonality among different samples. In most samples, the degree of physical pain has a similar trend and degree range of positive or negative influence on the prediction of anxious risk.

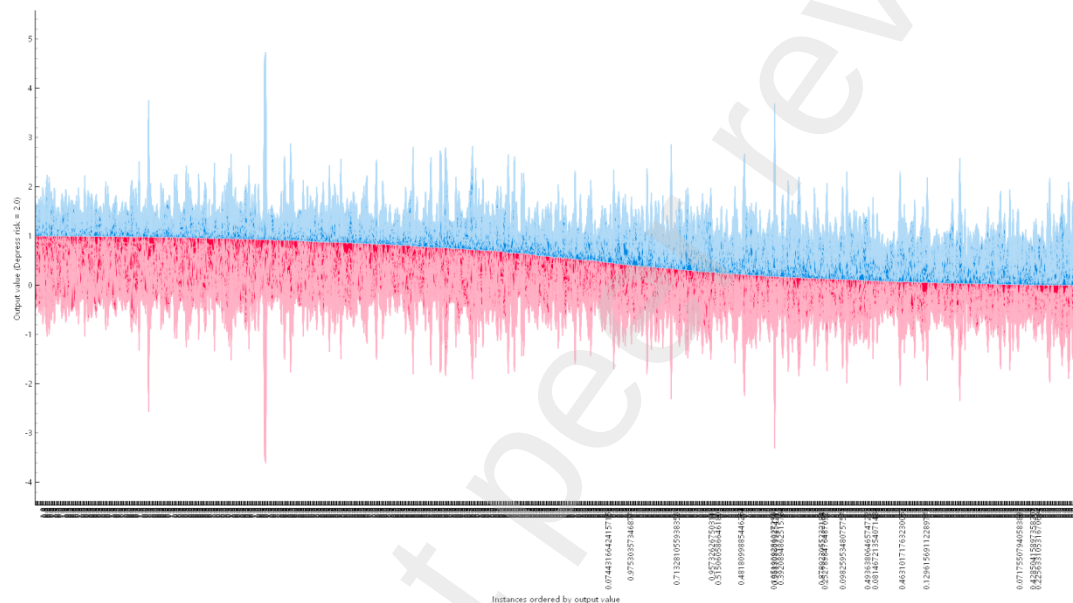


Figure 8. LIME values for body pain level on anxiety risk.

4. Discussion

In this study, by integrating the bio-psycho-social model framework with stacking ensemble learning technology, a prediction model for the anxious risk of elderly women receiving home-based care was constructed, revealing the nonlinear influence of multi-dimensional features on anxious symptoms.

The Stacking-SET ensemble model ($AUC = 0.818$) constructed in this study significantly outperformed single models (such as SVM-SET with an AUC of 0.777) in predicting the anxious risk of elderly women receiving home-based care, verifying the advantages of ensemble learning technology in scenarios of multi-dimensional feature interaction. After combining with data balancing techniques such as SMOTE-ENN+Tomek, the model performance improved by up to 20.9% (kNN-SET), indicating that high-quality data preprocessing plays a crucial role in alleviating the prevalent class imbalance problem in medical data (Maharana et al., 2022). The SHAP interpretability analysis showed that physiological factors such as the degree of body pain (with the highest SHAP value), sleep duration, and self-rated health assessment are core predictive variables. Meanwhile, environmental and sensory factors such as air quality satisfaction and hearing problems were revealed as important contributing factors for the first time, providing a new theoretical perspective for traditional biomedical models (Hao et al., 2024; Park et al., 2024).

The SHAP analysis identified key features such as the degree of body pain (with the highest SHAP value), sleep duration, and self-rated health assessment, which are consistent with existing

studies, indicating that physical health status is a core driving factor of anxious risk (Rahmati et al., 2024). It is worth noting that environmental and sensory factors such as air quality satisfaction and hearing problems ranked among the top 10 features, suggesting the potential impact of environmental adaptability on the mental health of women receiving home-based care. This provides a new perspective for traditional biomedical models.

The results of this study provide practical tools and intervention directions for clinicians, nurses, community workers, and other groups as follows. In terms of optimizing early screening tools, the model can help doctors quickly identify high-risk individuals by integrating multi-dimensional data (such as chronic diseases, ADL functions, social support, etc.) from electronic health records. For example, for elderly women with a body pain score ≥ 3 and a sleep duration ≤ 5 hours, it is recommended to prioritize psychological assessment.

In the dimension of precise intervention targets, community workers can design hierarchical intervention strategies based on the results of SHAP analysis. For example, for the environmentally sensitive group (with low air quality satisfaction), they can cooperate with environmental departments to optimize living conditions (Chen et al., 2024). In addition, for groups with weak social support (low children satisfaction), the family-community linkage support mechanism can be strengthened (Żyrek et al., 2024). In terms of optimizing resource allocation, the urban-rural universality of the model provides a basis for formulating differentiated mental health service plans in different regions, which helps to solve the problem of uneven distribution of medical resources (Zhuang et al., 2025).

This study achieved the integration of the advantages of multiple algorithms through the Stacking ensemble framework, breaking through the limitation of the traditional Logistic regression model (AUC = 0.746) in capturing nonlinear relationships. Compared with recent studies, this model has the following innovative aspects. Firstly, the divergence of the ICE curves of the total number of chronic diseases and labor time indicates significant individual heterogeneity in the influence of anxious risk. This echoes the "dynamic threshold effect" theory proposed by Wang et al. (2024), suggesting that future research needs to conduct dynamic modeling in combination with personalized features.

Secondly, the correlation between air quality and mental health verifies the applicability of the exposome theory in environmental epidemiology (Chen et al., 2024), providing a new interdisciplinary research direction. Finally, the model in this paper integrates the base model and the meta-model through a hierarchical learning mechanism, achieving a balance between prediction performance and generalization ability. This strategy is comparable to the ensemble method used by Qi et al. (2025) in cardiovascular risk prediction, but this study further strengthens the interpretability analysis (SHAP + ICE), providing an example for the "white-boxing" of medical artificial intelligence.

This study has certain limitations. Firstly, the data relies on the cross-sectional data of CLHLS in 2008, and it is impossible to verify the long-term prediction ability of the model. Secondly, anxious symptoms are self-reported through the GAD -7 scale, and there may be recall bias or misjudgment of undiagnosed cases. In addition, feature selection is based on Lasso regression, and some weakly correlated but clinically significant variables (such as economic pressure) may be omitted.

Future research can enhance the clinical transformation value of the model from dimensions such as multimodal data fusion (such as using wearable devices to monitor sleep quality), dynamic prediction models (such as using recurrent neural networks to process longitudinal data), and cross-cultural verification (such as comparing the differences in feature weights among East Asian, European, and American populations) (Chen et al., 2024). In addition, we plan to further increase the verification with longitudinal data, incorporate multi-period follow-up data, and

evaluate the temporal stability of the model.

Secondly, increase the verification with external data and combine objective biomarkers (such as inflammatory factors) to improve prediction accuracy. In addition, optimize the interpretability of influencing variables, and further analyze the feature interaction mechanism through tools to provide a basis for precise intervention. Finally, develop a simple assessment model suitable for the above-mentioned population and verify its applicability among other East Asian populations to promote the transformation of research results. Overall, this study provides an interpretable tool for the early identification of anxious risk in elderly women receiving home-based care, emphasizes the necessity of multi-dimensional interventions in the physiological, psychological, and social aspects, and has important practical value for optimizing community mental health services.

5. Conclusions

This study constructed a prediction model for the anxious risk of elderly women receiving home-based care based on Stacking ensemble learning. Using the cohort data of CLHLS in 2008, variables were defined in combination with the bio-psycho-social model. After feature screening by LASSO regression and data processing with multiple data balancing techniques, the model was constructed, and the model was explained through SHAP analysis.

The study found that there is a certain prevalence of anxiety in this group. The Stacking-SET ensemble model performed excellently in predicting anxious risk, outperforming single models, and combining with data balancing techniques can significantly improve the model performance. The model revealed multiple key predictive factors such as the degree of body pain, sleep duration, and self-rated health assessment, providing a basis for early screening, precise intervention, and resource allocation.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics approval and consent to participate

The studies involving humans were approved by The Biomedical Ethics Review Committee of Peking University (Beijing, China). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Supplementary material

The Supplementary material for this article can be found online.

References

- Ai F., Li E., Ji Q., Zhang H., 2024. Construction of a machine learning-based risk prediction model for anxiety in middle-aged and elderly hypertensive people in China: a longitudinal study. *Front Psychiatry*. 15, 1398596. doi: 10.3389/fpsy.2024.1398596.
- AlMahadin G, Lotfi A, Carthy MM, Breedon P., 2021. Task-Oriented Intelligent Solution to Measure Parkinson's Disease Tremor Severity. *J Healthc Eng*. 9624386. doi:

10.1155/2021/9624386.

- Alsinglawi B, Alshari O, Alorjani M, Mubin O, Alnajjar F, Novoa M, Darwish O.,2022. An explainable machine learning framework for lung cancer hospital length of stay prediction. *Sci Rep.* 12, 607. doi: 10.1038/s41598-021-04608-7.
- An R, Gao Y, Huang X, Yang Y, Yang C, Wan Q., 2024. Predictors of progression from subjective cognitive decline to objective cognitive impairment: A systematic review and meta-analysis of longitudinal studies. *Int J Nurs Stud.* 149, 104629. doi: 10.1016/j.ijnurstu.2023.104629.
- Chen Q, Zhang X, Wang Z, et al.,2024. Data-driven prediction of dimensionless quantities for semi-infinite target penetration by integrating machine-learning and feature selection methods. *Defence Technology*, 40,105 - 124.
- Chen T, Dai K, Wu H., 2024. Persistent organic pollutants exposure and risk of anxiety : A systematic review and meta-analysis. *Environ Res.* 263, 120160. doi: 10.1016/j.envres.2024.120160.
- Chen YX, Zheng J, Zhang XF., 2024. Association analysis between organophosphorus flame retardants exposure and the risk of anxiety : Data from NHANES 2017-2008. *J Affect Disord.* 355, 385-391. doi: 10.1016/j.jad.2024.04.004.
- Cui, L., Li, S., Wang, S., Wu, X. F., Liu, Y., Yu, W., Wang, Y., Tang, Y., Xia, M. S., & Li, B. M., 2024. Major anxious disorder: hypothesis, mechanism, prevention and treatment. *Signal Transduction and Targeted Therapy*, 9, 30. <https://doi.org/10.1038/s41392-024-01738-y>.
- Delshad V, Stueck M, Ebadi A, Bidzan M, Khankeh H., 2022. Bio-psycho-social health assessment in prehospital emergency technicians: A systematic review. *J Educ Health Promot.* 11, 41. doi: 10.4103/jehp.jehp_498_21.
- Diaz-Gallo LM, Brynedal B, Westerlind H, Sandberg R, Ramsköld D., 2021. Understanding interactions between risk factors, and assessing the utility of the additive and multiplicative models through simulations. *PLoS One.* 16, e0250282. doi: 10.1371/journal.pone.0250282.
- Ehtemam H, Sadeghi Esfahlani S, Sanaei A, Ghaemi MM, Hajesmaeel-Gohari S, Rahimisadegh R, Bahaadinbeigy K, Ghasemian F, Shirvani H., 2024. Role of machine learning algorithms in suicide risk prediction: a systematic review-meta analysis of clinical studies. *BMC Med Inform Decis Mak.* 24, 138. doi: 10.1186/s12911-024-02524-0.
- Guo J, Wang J, Sun W, Liu X., 2022. The advances of post-stroke anxiety : 2021 update. *J Neurol.* 269, 1236 - 1249. doi: 10.1007/s00415-021-10597-4.
- Han Y, Wang S., 2023. Disability risk prediction model based on machine learning among Chinese healthy older adults: results from the China Health and Retirement Longitudinal Study. *Front Public Health.* 11, 1271595. doi: 10.3389/fpubh.2023.1271595.
- Hao Y, Xu L, Peng M, Yang Z, Wang W, Meng F., 2024. Synergistic air pollution exposure elevates anxiety risk: A cohort study. *Environ Sci Ecotechnol.* 23, 100515. doi: 10.1016/j.ese.2024.100515.
- Kim JY, Lim MH.,2021. Psychological factors to predict chronic diarrhea and constipation in Korean high school students. *Medicine (Baltimore).* 100, e26442. doi: 10.1097/MD.00000000000026442.
- Li E, Ai F, Tian Q, Yang H, Tang P, Guo B., 2025. Develop and validate machine learning models to predict the risk of anxious symptoms in older adults with cognitive impairment. *BMC Psychiatry.* 25, 219. doi: 10.1186/s12888-025-06657-y.
- Maharana, K., Mondal, S., & Nemade, B., 2022. A review: Data pre-processing and data augmentation techniques. *Global Transitions Proceedings*, 3, 91–99. <https://doi.org/10.1016/j.gltp.2022.04.020>.
- Nguyen T, Mengersen K, Sous D, Liqueur B., 2023. SMOTE-CD: SMOTE for compositional data.

- Ou H, Yao Y, He Y., 2024. Missing Data Imputation Method Combining Random Forest and Generative Adversarial Imputation Network. *Sensors (Basel)*. 24, 1112. doi: 10.3390/s24041112.
- Park W, Jang H, Ko J, Sohn J, Noh Y, Kim SY, Koh SB, Kim C, Cho J., 2024. Physical Activity-Induced Modification of the Association of Long-Term Air Pollution Exposure with the Risk of Anxiety in Older Adults. *Yonsei Med J*. 65, 227-233. doi: 10.3349/ymj.2023.0292.
- Qiu Y, Ma X., 2024. Using machine learning models to identify the risk of anxiety in middle-aged and older adults with frequent and infrequent nicotine use: A cross-sectional study. *J Affect Disord*. 367, 554 - 561. doi: 10.1016/j.jad.2024.08.185.
- Qi X, Wang S, Fang C, Jia J, Lin L, Yuan T., 2025. Machine learning and SHAP value interpretation for predicting comorbidity of cardiovascular disease and cancer with dietary antioxidants. *Redox Biol*. 79, 103470. doi: 10.1016/j.redox.2024.103470.
- Rahmati M, Lee S, Yon DK, Lee SW, Udeh R, McEvoy M, Oh H, Butler L, Keyes H, Barnett Y, Koyanagi A, Shin JI, Smith L., 2024. Physical activity and prevention of mental health complications: An umbrella review. *Neurosci Biobehav Rev*. 160, 105641. doi: 10.1016/j.neubiorev.2024.105641.
- Tan J, Ma C, Zhu C, Wang Y, Zou X, Li H, Li J, He Y, Wu C., 2022. Prediction models for anxiety risk among older adults: systematic review and critical appraisal. *Ageing Res Rev*. 101803. doi: 10.1016/j.arr.2022.101803.
- Tan J, Ma C, Zhu C, Wang Y, Zou X, Li H, Li J, He Y, Wu C., 2023. Prediction models for anxiety risk among older adults: systematic review and critical appraisal. *Ageing Res Rev*. 101803. doi: 10.1016/j.arr.2022.101803.
- Tang G, Qi L, Sun Z, Liu J, Lv Z, Chen L, Huang B, Zhu S, Liu Y, Li Y., 2021. Evaluation and analysis of incidence and risk factors of lower extremity venous thrombosis after urologic surgeries: A prospective two-center cohort study using LASSO-logistic regression. *Int J Surg*. 105948. doi: 10.1016/j.ijssu.2021.105948.
- Vu T, Dawadi R, Yamamoto M, Tay JT, Watanabe N, Kuriya Y, Oya A, Tran PNH, Araki M., 2025. Prediction of anxious disorder using machine learning approaches: findings from the NHANES. *BMC Med Inform Decis Mak*. 25, 83. doi: 10.1186/s12911-025-02903-1.
- Wang H, Chen G, Sun D, Ma Y., 2024. The threshold effect of triglyceride glucose index on diabetic kidney disease risk in patients with type 2 diabetes: unveiling a non-linear association. *Front Endocrinol (Lausanne)*. 15, 1411486. doi: 10.3389/fendo.2024.1411486.
- Wang K, Tian J, Zheng C, Yang H, Ren J, Liu Y, Han Q, Zhang Y., 2021. Interpretable prediction of 3-year all-cause mortality in patients with heart failure caused by coronary heart disease based on machine learning and SHAP. *Comput Biol Med*. 137, 104813. doi: 10.1016/j.combiomed.2021.104813.
- Wang L, Harris R, Simoni JM, Yue Q, Fu J, Zheng H, Ning Z, Xavier Hall CD, Burns PA, Wong FY., 2023. Health Service Utilization and Its Associations with Anxiety and Sexual Risk Behaviors Among Transgender Women in Shanghai, China. *Transgend Health*. 8, 516 - 525. doi: 10.1089/trgh.2021.0009.
- Wang Q, Fan K, Li P., 2022. Effect of the Use of Home and Community Care Services on the Multidimensional Health of Older Adults. *International Journal of Environmental Research and Public Health*, 19, 15402. doi: 10.3390/ijerph192215402
- Wang W, Liu Y, Ji D, Xie K, Yang Y, Zhu X, Feng Z, Guo H, Wang B., 2024. The association between functional disability and anxious symptoms among older adults: Findings from the China Health and Retirement Longitudinal Study (CLHLS). *J Affect Disord*. 351, 518 - 526.

doi: 10.1016/j.jad.2024.01.256.

- Wang X, Hu J, Wu D., 2022. Risk factors for frailty in older adults. *Medicine (Baltimore)*. 101, e30169. doi: 10.1097/MD.00000000000030169.
- Wang Y, Zhang Y, Ni B, Jiang Y, Ouyang Y., 2023. Development and validation of a anxiety risk prediction nomogram for US Adults with hypertension, based on NHANES 2007 - 2008. *PLoS One*. 18, e0284113. doi: 10.1371/journal.pone.0284113.
- Wei J, Hou R, Zhang X, Xu H, Xie L, Chandrasekar EK, Ying M, Goodman M., 2019. The association of late-life anxiety with all-cause and cardiovascular mortality among community-dwelling older adults: systematic review and meta-analysis. *Br J Psychiatry*. 215, 449 - 455. doi: 10.1192/bjp.2019.74.
- Zheng Y, Zhang C, Liu Y., 2024. Risk prediction models of anxiety in older adults with chronic diseases. *J Affect Disord*. 359, 182 - 188. doi: 10.1016/j.jad.2024.05.078.
- Zhuang X, Chan CP, Yang X., 2025. A network comparison analysis of socio-ecological protective and risk factors of anxiety between Chinese urban and rural adolescents. *Soc Sci Med*. 365, 117628. doi: 10.1016/j.socscimed.2024.117628.
- Żyrek J, Klimek M, Apanasewicz A, Ciochoń A, Danel DP, Marcinkowska UM, Mijas M, Ziomkiewicz A, Galbarczyk A., 2024. Social support during pregnancy and the risk of postpartum anxiety in Polish women: A prospective study. *Sci Rep*. 14, 6906. doi: 10.1038/s41598-024-57477-1.